Questions

Q1.

Diagram 1 shows a glass prism which can be used to turn an image the right way up.

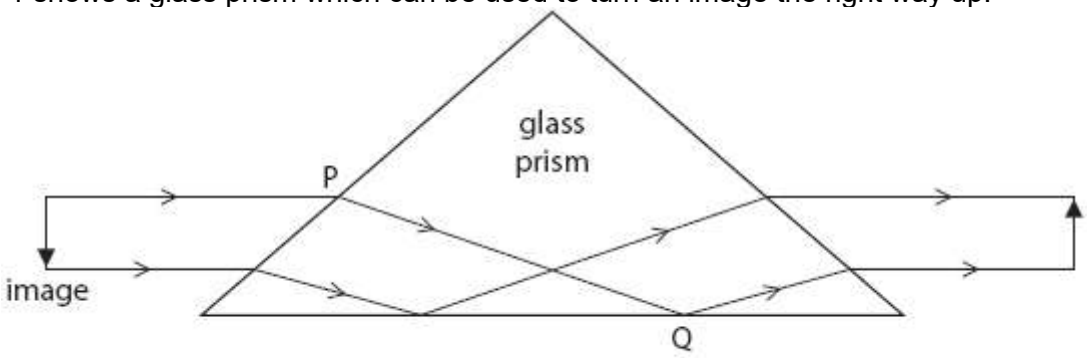

Diagram 1

(i) In diagram 1, total internal reflection occurs at Q.
Explain why total internal reflection occurs at Q.

(2)

.....

.....

.....

.....

(ii) The way in which the light changes direction at P is shown in diagram 2.
Mark on the diagram (*i*) for the angle of incidence and (*r*) for the angle of refraction for the ray of light shown.

(2)

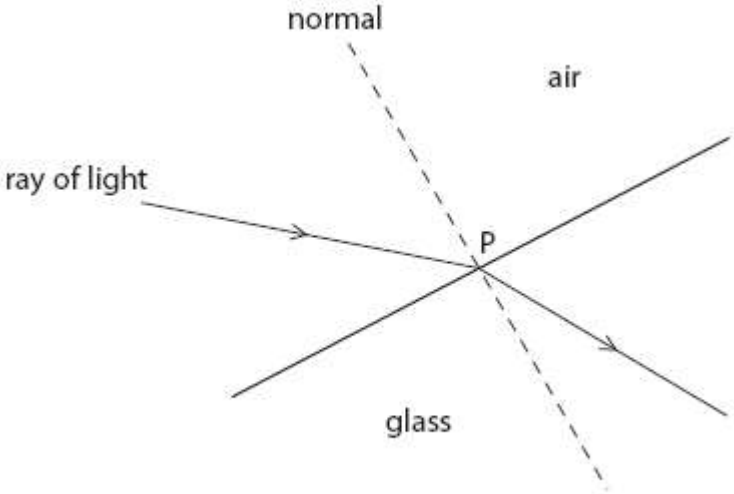

Diagram 2

(iii) Which of these is correct for the light as it enters the prism at P?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** frequency decreases

☐ **B** frequency increases

☐ **C** speed decreases

☐ **D** speed increases

Q2.

A student investigates what happens when light travels from air to glass.

Figure 15 shows some of the apparatus used in the investigation.

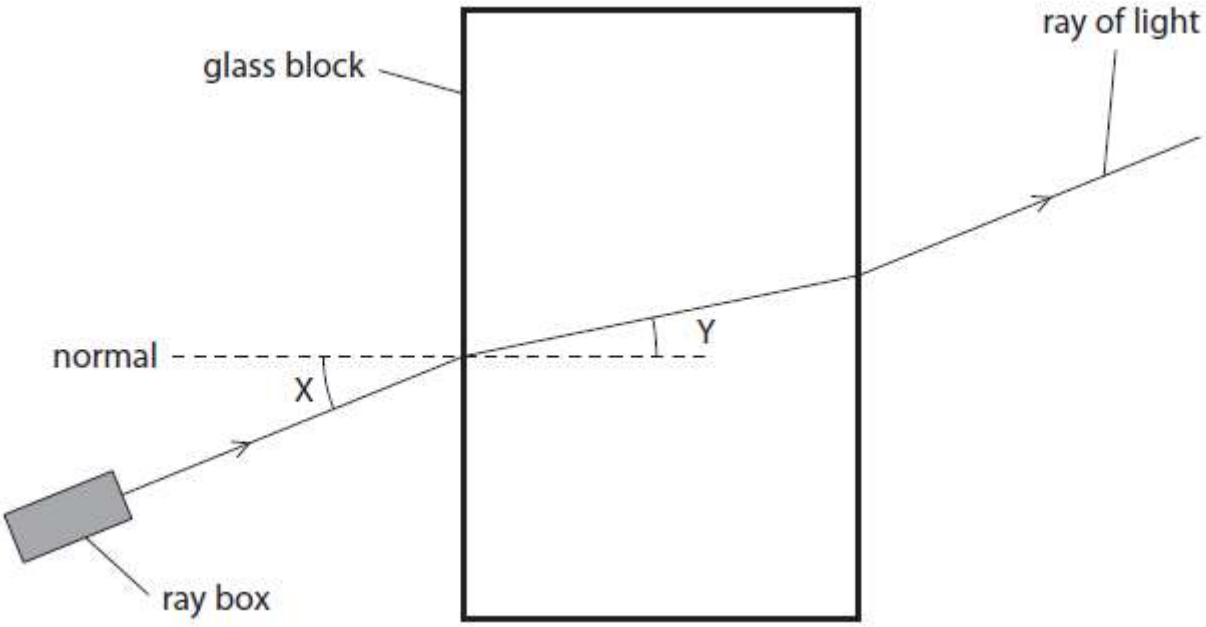

Figure 15

(i) In Figure 15, angle Y is the angle of

(1)

- ☐ **A** deflection
- ☐ **B** incidence
- ☐ **C** reflection
- ☐ **D** refraction

(ii) Figure 16 is a graph of the student's results.

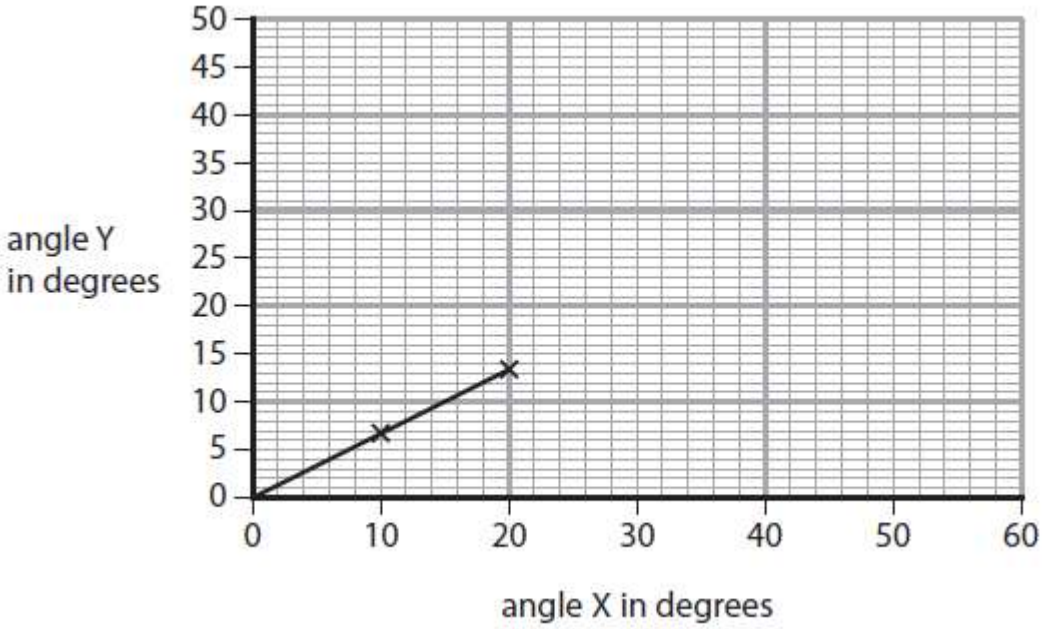

Figure 16

Use the graph to calculate a value for

$$\frac{\text{angle Y}}{\text{angle X}}$$

(2)

$\frac{\text{angle Y}}{\text{angle X}} =$ .....

(iii) The student concludes that angle Y is directly proportional to angle X.

Explain what the student must do to test this conclusion in more detail.

(3)

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.....

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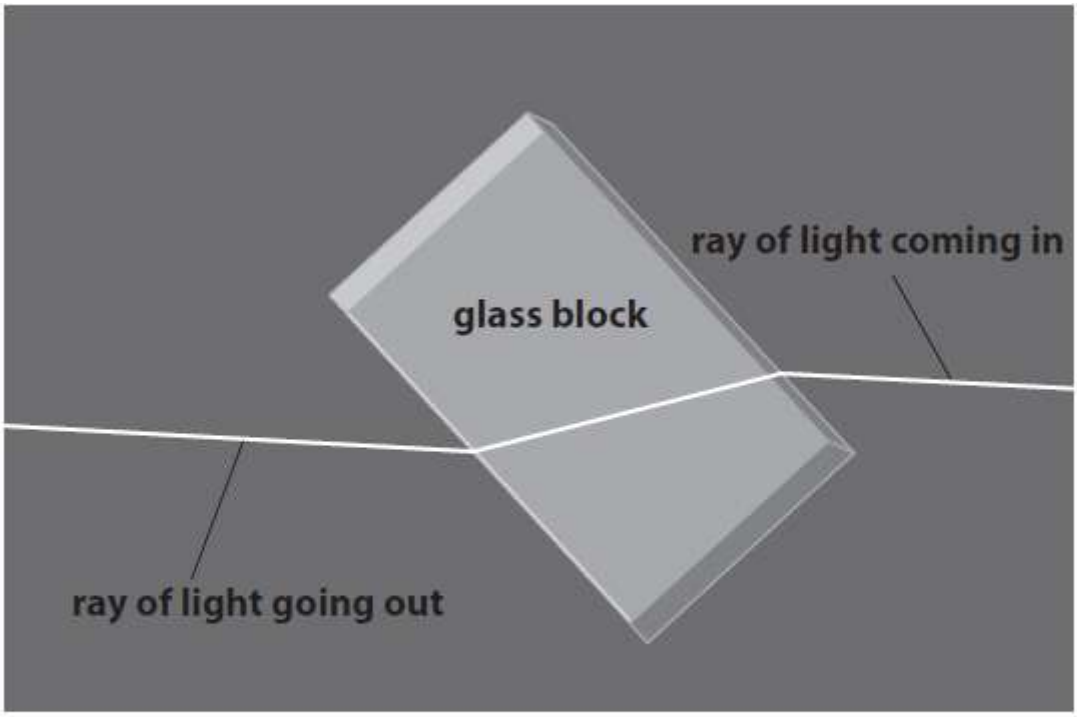
.....

(Total for question = 6 marks)

Q3.

Figure 2 is a diagram of a glass block being used to investigate the refraction of light.



(Source: © ajizai/docstoc.com)

Figure 2

The angle of incidence was varied and the angle of refraction was measured each time.

Figure 3 shows the results in a table.

angle of incidence in degrees	angle of refraction in degrees
20	13
30	20
40	25
50	31
60	35
70	39
80	41

Figure 3

Figure 4 shows a graph of these results.

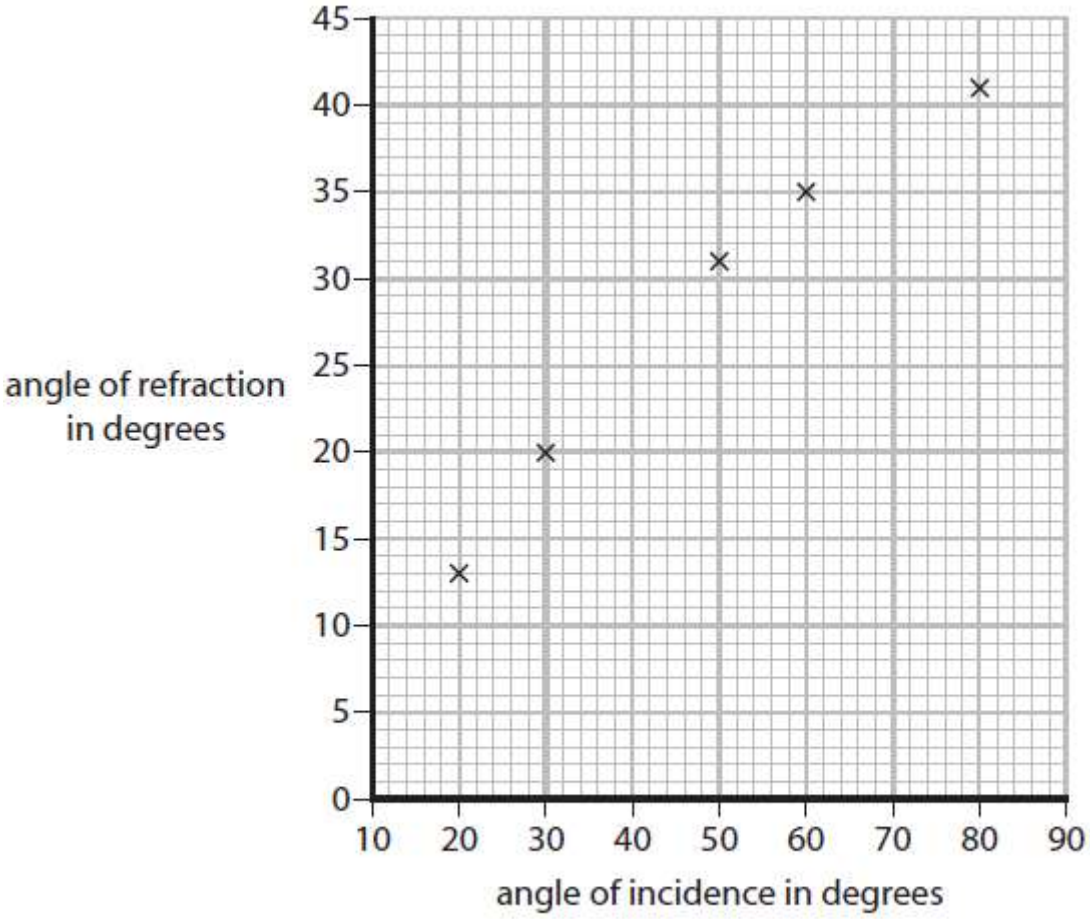


Figure 4

(i) The two shaded rows in Figure 3 are not plotted on the graph in Figure 4.
Plot these missing results on the graph in Figure 4.

(2)

(ii) Draw a curve of best fit through the points on the graph in Figure 4.

(1)

(i) Using Figure 4, describe the relationship between the angle of refraction and the angle of incidence.

(2)

.....

.....

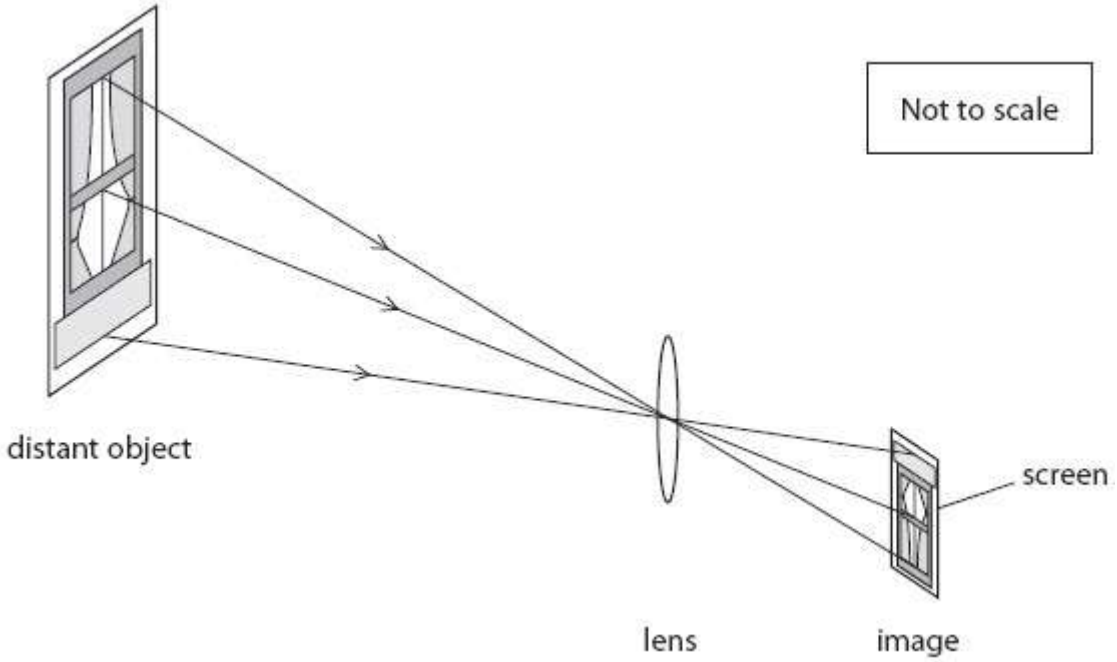
.....

.....

(Total for question = 5 marks)

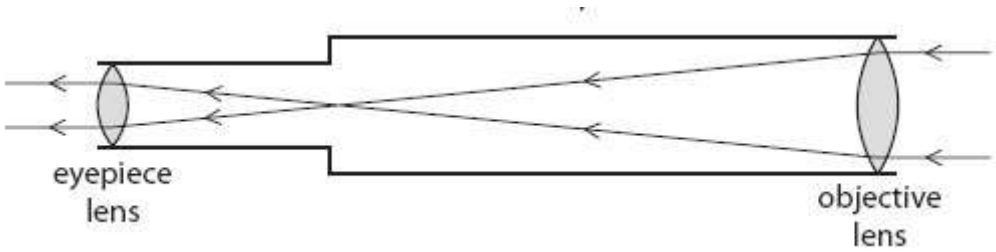
Q4.

A lens can be used to produce a clear image of a distant object on a screen.

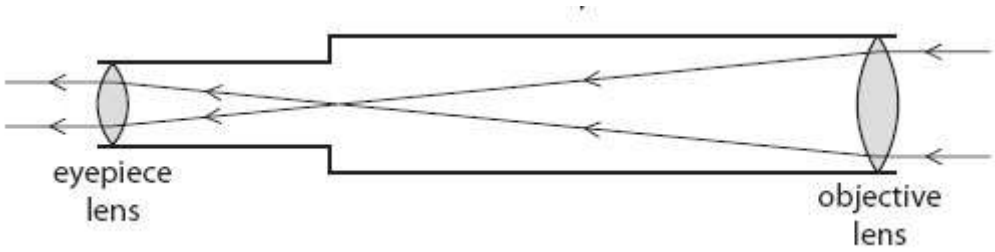


(a) (i) Complete the sentence by putting a cross (■) in the box next to your answer.
The image produced is real because it is

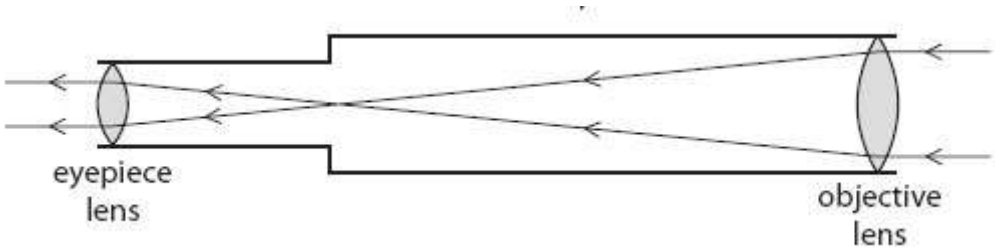
(1)



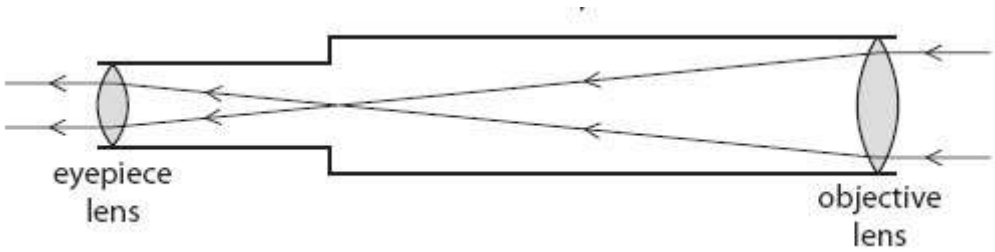
A in focus



B magnified



C on a screen



D smaller

(ii) Describe how to measure the focal length of the lens.

(2)

.....

.....

.....

.....

(b) The diagram shows a simple telescope which uses two lenses to look at stars.

(i) Explain what the eyepiece lens does.

(2)

.....

.....

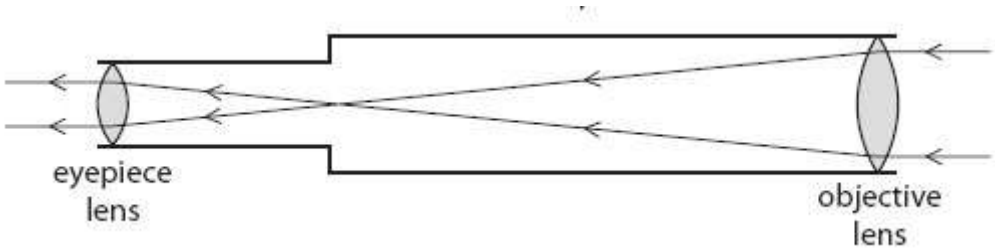
.....

.....

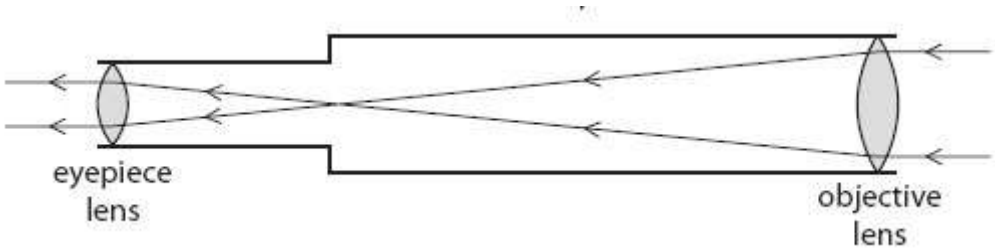
(ii) Complete the sentence by putting a cross (■) in the box next to your answer.

The light that travels from the stars transfers

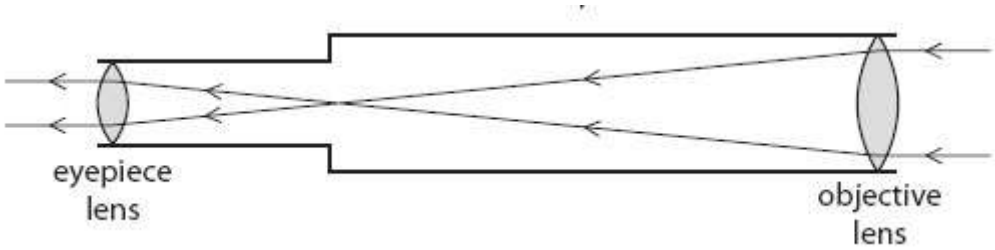
(1)



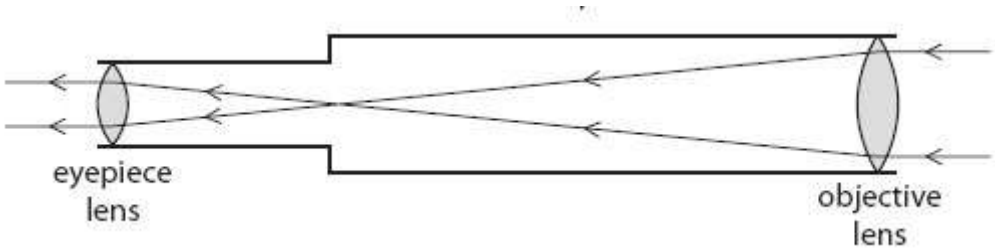
A charge



B energy



C mass



D matter

(c) Light and sound waves are produced at the same time by an explosion on Earth.

(i) The sound of the explosion is heard 1920 metres away 6.0 seconds after the explosion has happened.

Calculate the speed of sound in air.

(2)

.....

(ii) A scientist is standing a long way from the explosion.

Explain why he hears the explosion a few seconds after he sees it.

(2)

.....

.....

.....

.....

(Total for Question = 10 marks)

Q5.

This question is about reflection and refraction of light.

A converging lens has a focal length of 40 cm.
Calculate the power of this lens in dioptres.
Use the equation

$$\text{power in dioptres} = \frac{1}{\text{focal length in metres}}$$

(3)

power of the lens ..... dioptres

(Total for question = 3 marks)

Q6.

A lens has a focal length of 17 cm.

Calculate the power of the lens in dioptres.

Use the equation

$$\text{power (in dioptres)} = \frac{1}{\text{focal length (in metres)}}$$

(3)

power = ... dioptres

(Total for question = 3 marks)

Q7.

Calculate the power of a lens of focal length 17 cm.

Use the equation

$$\text{power (in dioptres)} = \frac{1}{\text{focal length (in metres)}}$$

Give the answer to 2 significant figures.

(3)

power = ... dioptres

(Total for question = 3 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Figure 1 shows how the visible spectrum of white light is shown on a screen.

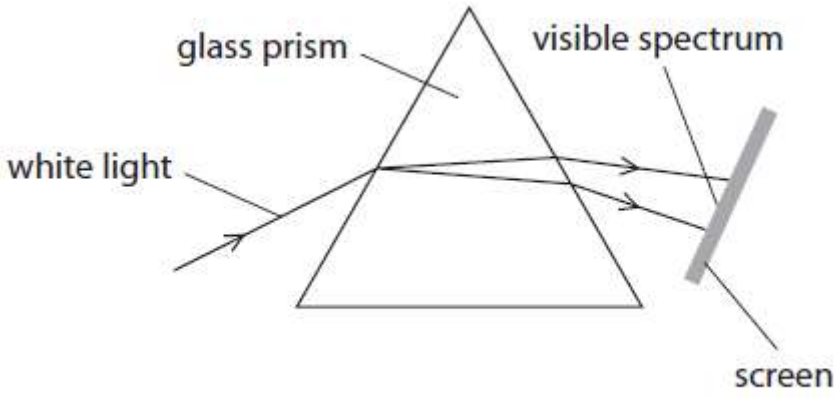


Figure 1

(i) Which of these is the best piece of equipment to produce the white light?

(1)

- ☐ **A** ray box
- ☐ **B** ruler
- ☐ **C** measuring cylinder
- ☐ **D** ammeter

(ii) Which colour is seen between yellow and blue in the spectrum on the screen?

(1)

- ☐ **A** red
- ☐ **B** orange
- ☐ **C** green
- ☐ **D** violet

(Total for question = 2 marks)

Q9.

A student uses a ray box to investigate two lenses, P and Q.

Figure 6 shows the student's drawings of rays of light before and after passing through the lenses.

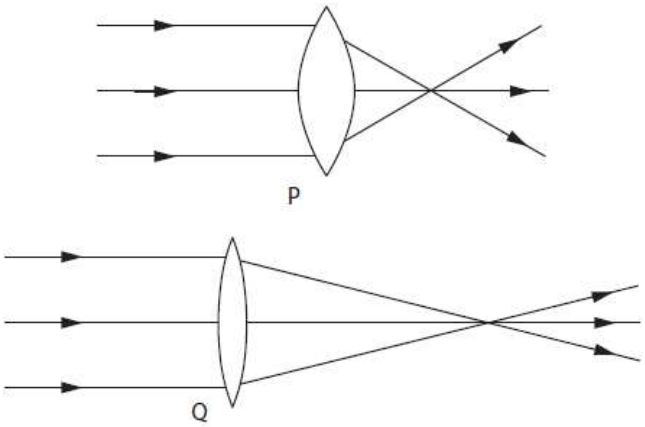


Figure 6

Complete the following sentences.

(2)

The focal length of lens P is ..... than the focal length of lens Q.

The power of lens P is ..... than the power of lens Q.

(Total for question = 2 marks)

Q10.

Figure 2 shows the main parts of the electromagnetic spectrum.

radio	microwaves	infrared	visible light	ultraviolet	x-rays	gamma rays

Figure 2

Complete the following sentences using information from Figure 2. Each part of the electromagnetic spectrum may be used once, more than once or not at all.

(i) The part of the electromagnetic spectrum used to detect broken bones is

(1)

.....

(ii) The part of the electromagnetic spectrum used in thermal imaging is

(1)

.....

(iii) The part of the electromagnetic spectrum that
- is used to cook food

AND
- has a shorter wavelength than microwaves is

(1)

.....

(iv) The part of the electromagnetic spectrum that
- is used to sterilise medical equipment

AND
- has a shorter wavelength than x-rays is

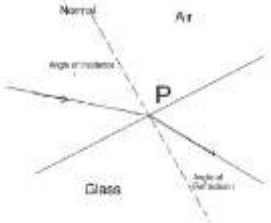
(1)

.....

(Total for question = 4 marks)

Mark Scheme

Q1.

	Answer	Acceptable answers	Mark
(i)	An explanation linking: Angle (of incidence) in glass (1) greater than critical angle / 42° (1)	Angle in air cannot be greater than 90° for 1 mark Glass has a higher refractive index than air for 1 mark	**(2)**
(ii)	 angle i from normal in air (1) angle r from normal in glass (1)	accept for 1 mark angle i in air <u>and</u> angle r in glass/ <u>both</u> angles measured from normal	**(2)**
(iii)	☒ **C** speed decreases		**(1)**

Q2.

Question Number	Answer	Mark
(i)	**D** refraction **The only correct answer is D** *A 'deflection' is an incorrect distracting description* *B 'incidence' is incorrect, that would be angle X* *C 'reflection' is incorrect, no reflection being shown in the diagram*	**(1)** AO 1 1

Question Number	Answer	Additional guidance	Mark
(ii)	any pair of coordinates selected from the line (1) in range → 0.6(0) to 0.7(0) (1)	e.g. 20 and (13 or 14) or 10 and (6 or 7) ignore any units given award full marks for a correct answer without working	**(2)** AO 2 1

Question Number	Answer	Additional guidance	Mark
(iii)	an explanation linking: repeat (1) different angles / more values of X (1) for larger angles / values of X (1)	 allow 'more measurements' / 'repeat experiment' / collect more data > 20°	(3) AO 3 3a

Q3.

Q4.

	Answer	Acceptable answers	Mark
(a)(i)	☒ C on a screen		(1)
(a)(ii)	Explanation linking the following:- (measure) distance / length / from / line (1) lens to image / screen / focal point (1)	how far away use ruler / tape measure measure u(object distance) and v(image distance) calculate focal length using 1/u + 1/v = 1/f (measure) from lens to image (2)	(2)
(b)(i)	A description including the following: • magnifies • the image • refracts the light	brings nearer / zooms in / looks closer / makes bigger / enlarges intermediate / real image	(2)
(b)(ii)	☒ B energy		(1)

(c)(i)	Substitution v = 1920/6.0 (1) Evaluation (1) 320 (m/s)	300 (m/s) give full marks for correct answer, no working (2) Power of 10 error max. 1 mark.	**(2)**
(c)(ii)	Suggestions including the following: • recognition of any difference in speed / velocity (1) • correct difference in speed (1)	e.g. sound travels faster / quicker than light (1) c>v / v<c / c>320 (m/s) Light travels (much) faster (2) RA **Ignore** 'sound takes longer' or other references to time.	**(2)**

Q5.

	Answer	Additional guidance	Mark
	substitution (1) evaluation without correct unit change (1) evaluation including unit change (1) 2.5 (dioptres)	(power) = $\frac{1}{0.40}$ allow 4 to any power of 10 in the substitution allow 2.5 to any power of 10 (dioptres) 2.5 to any other power of 10 scores 2 marks award full marks for the correct answer without working	**(3)** **AO2**

Q6.

Q7.

Question Number	Answer	Additional guidance	Mark
	substitution (1) $\frac{1}{17}$ or 0.17 evaluation (1) 5.882... evaluation to 2sf (1) 5.9 (any answer to 2 sf for this mark, but not if wrong rounding is seen)	lose this mark if there is **any other power of 10 error** but then apply ecf (e.g. 1/1.7) so 0.588 then = 1 mark for that ecf independent mark award full marks for the correct answer without working	**(3)** AO 2 1

Q8.

Question Number	Answer	Mark
(i)	**A ray box** ***B*** *is not correct because a ruler does not produce a beam of white light* ***C*** *is not correct because a measuring cylinder does not produce a beam of white light* ***D*** *is not correct because an ammeter does not produce a beam of white light*	**(1)** **AO1**

Question Number	Answer	Mark
(ii)	**C green** ***A*** *is not correct because red appears at the start of the spectrum* ***B*** *is not correct because orange appears in the middle of the spectrum* ***D*** *is not correct because violet appears at the end of the spectrum*	**(1)** **AO1**

Q9.

Question number	Answer	Additional guidance	Mark
	shorter / less / smaller (1) greater / larger / more / stronger (1)	lower higher	**2** **AO1.1**

Q10.

	Answer	Additional guidance	Mark
(i)	x-ray(s)	allow X x no mark if more than one wave given e.g. x-rays and gamma rays scores 0	**(1)** **AO1**

	Answer	Additional guidance	Mark
(ii)	infrared	allow any recognisable spelling IR ir no mark if more than one wave given e.g. infrared and gamma rays scores 0	**(1)** **AO1**

	Answer	Additional guidance	Mark
(iii)	infrared	allow any recognisable spelling IR ir no mark if more than one wave given e.g. infrared and gamma rays scores 0	**(1)** **AO1**

	Answer	Additional guidance	Mark
(iv)	gamma (rays)	allow any recognisable spelling γ no mark if more than one wave given e.g. gamma rays and UV scores 0	**(1)** **AO1**